

Predictors of Mental Health Treatment-Seeking Behavior in the Technology Sector

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Abstract--- *This study aims to investigate the key factors influencing mental health treatment-seeking behavior among employees in the tech sector. Using survey data consisting of 1250 observations, we analyzed the relationships between the target variable ("treatment") and various demographic, workplace, and mental health-related predictors. Methodologically, the first phase included nominal and ordinal association tests, effect size measurements (Odds Ratio, Cramér's V), and the Cochran-Mantel-Haenszel (CMH) test on three-way tables to control for confounding effects. In the second phase, we built logistic regression, regularization, and tree-based machine learning algorithms to predict the likelihood of seeking treatment. Among the evaluated models, Elastic Net provided the most robust and stable performance on the test set (ROC-AUC = 0.8067) by effectively balancing model complexity. The analysis shows that a family history of mental illness, work interference caused by mental health conditions, and employer-provided benefits are the strongest predictors of treatment-seeking behavior. These findings emphasize how critical workplace policies and support systems are for employees utilizing mental health services.*

I. INTRODUCTION & DATA SOURCE

A. Background

Employees in the tech sector often work under demanding conditions, including tight deadlines, rapidly changing technologies, and high performance expectations. These conditions may contribute to increased psychological stress in the workplace. As a result, issues such as burnout, anxiety, and depression are frequently discussed as important concerns within the technology workforce. Understanding the factors associated with treatment-seeking behavior may help organizations better support employee well-being while also reducing the negative effects of mental health problems in the workplace. Many people still hesitate to get treatment because they fear being stigmatized, worry about their careers, or simply lack corporate support. Because of this, it is highly important to categorically analyze the factors that drive treatment decisions and to create reliable predictive models. Insights obtained from such analyses may assist organizations in improving workplace policies and increasing awareness of available mental health resources. The main objective of this study is to determine the relationships

between employees' treatment-seeking behavior and their demographics, workplace conditions, and personal perceptions, and to build a statistical model that accurately predicts this behavior.

B. Dataset & Problem Definition

The dataset used in this research is based on a global survey conducted in 2014 to measure mental health status, perceptions, and corporate support mechanisms in the tech sector. The raw dataset initially contained 1259 observations and 27 variables. After applying data cleaning and age-filtering procedures, the final analysis and modeling processes were carried out on 1250 observations.

Target (Dependent) Variable: The focal point of the study is the treatment variable. It has a binary structure ("Yes" or "No") indicating whether the individual has sought professional treatment for a mental health condition.

Independent Variables: The predictors in the dataset are classified into three main groups to reflect their explanatory power and nature:

Demographic Variables: This includes Age (and the derived AgeGroup) which shows the age structure of the participants, along with the standardized Gender variable.

Workplace Conditions and Policies: This group consists of CompanySize (collapsed from the no_employees variable), remote work status (remote_work), whether the individual is self-employed (self_employed), whether the employer is a tech company (tech_company), employer-provided health insurance benefits (benefits), awareness of care options (care_options), corporate wellness programs (wellness_program), resources for seeking help (seek_help), protection of anonymity (anonymity), and the ease of taking medical leave for mental health issues (leave).

Mental Health and Perception Variables: This includes family history of mental illness (family_history), the ordinal variable measuring how much mental health issues interfere with work performance (work_interfere), the perceived potential negative career consequences of discussing mental or physical health issues at work

(mental_health_consequence, phys_health_consequence), openness to discussing these topics with coworkers and supervisors (coworkers, supervisor), willingness to share this information in job interviews (mental_health_interview, phys_health_interview), the perception of whether the employer takes mental health as seriously as physical health (mental_vs_physical), and whether the employee has observed negative consequences for coworkers with mental health issues (obs_consequence).

II. DATA CLEANING & PREPROCESSING

A. Data Cleaning

The raw dataset initially contained 1259 observations. To improve data quality, illogical age entries were filtered out, restricting the analysis to adult individuals between the ages of 18 and 70 (n=1250). Variables with excessive missing data or those in a free-text format unsuitable for analytical modeling, such as Timestamp, state, and comments, were removed to reduce noise. Additionally, the open-ended Gender variable was carefully examined. The numerous spelling variations (e.g., "male", "cis male", "woman", etc.) were cleaned using string manipulation techniques and standardized into three main categories: "Male", "Female", and "Other".

B. Missing Data Handling

An inspection of the variables revealed missing values, particularly in the work_interfere and self-employed columns. For practical modeling purposes, missing observations were treated as non-informative, and the missingness pattern was assumed not to systematically depend on the target variable. To strictly prevent data leakage during the machine learning phase, the dataset was first split into training (70%) and testing (30%) subsets using stratified sampling before any imputation took place. After isolating the test data, the missing observations were filled using Mode Imputation calculated solely from the training set. The distribution of missing data after the cleaning step is shown in Figure 1.

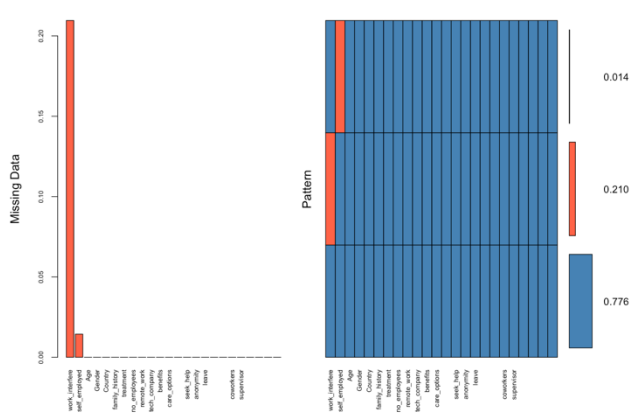
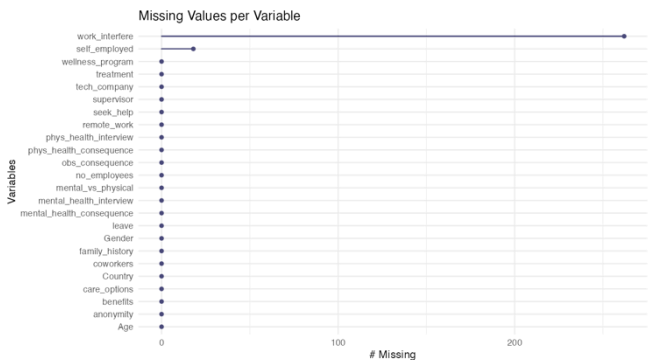


Figure 1: Missing Values Analysis and Imputation Pattern

III. EXPLORATORY DATA ANALYSIS (EDA)

In this section, we explore the structural characteristics, distributions, and relationships of the variables with the target variable in the cleaned and preprocessed dataset, using descriptive statistics and visualization techniques. The analysis is structured into two main stages: univariate and bivariate examinations.

A. Univariate Analysis

Before moving on to modeling and inference, examining the marginal distributions of the variables is a critical step for assessing data quality and checking model assumptions. To start, we looked at our dependent (target) variable, treatment, which indicates whether an individual sought mental health treatment.

Out of 1250 observations, 50.48% (631 people) received treatment ("Yes"), while 49.52% (619 people) did not ("No"). Because the target variable has an almost perfectly balanced distribution, the logistic regression and machine learning algorithms we build later will not suffer from class bias. This also means we do not need to apply artificial balancing techniques like SMOTE, oversampling, or undersampling.

Looking at the univariate distributions of the independent predictors, work_interfere (the degree to which mental health interferes with work) stands out as a clear reflection of workplace dynamics in the tech sector. A large portion of the participants stated that mental health issues affect their work processes at least "Sometimes" or "Rarely". Similarly, for variables measuring corporate support mechanisms like benefits and care_options, there is a noticeably high frequency of "Don't know" responses. This points to a lack of transparency or suggests that employees across the industry do not fully grasp their corporate policies.

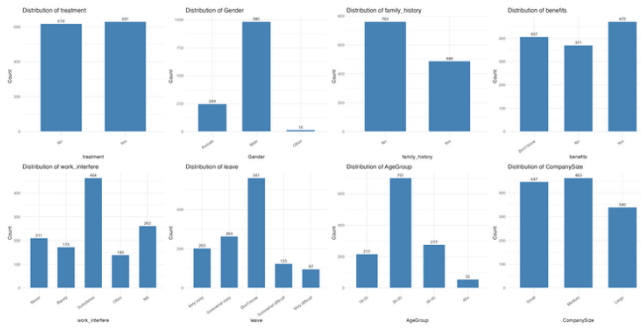


Figure 2: Univariate Frequency Distributions of the Target Variable and Key Predictors

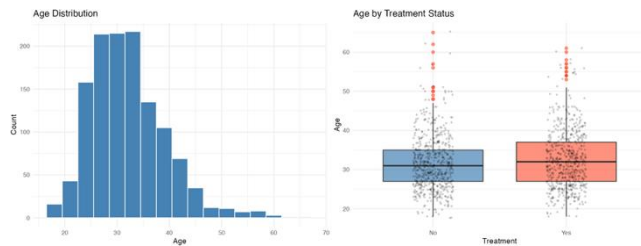


Figure 3: Histogram and Distribution Characteristics of Age

Looking at the demographics, the age distribution (Age) is heavily concentrated in the young-to-middle-aged group, mostly between 25 and 35 years old (Figure 3).

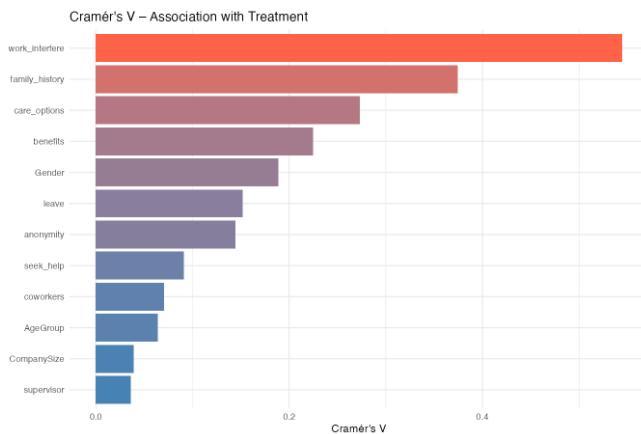


Figure 4: Cramér's V Correlation Strengths for Categorical Predictors

B. Bivariate/Multivariate Analysis

After completing the univariate checks, we created bivariate cross-tabulations and calculated proportions to capture the raw relationships and trends between the independent predictors and the target variable (treatment).

We observed that the factor creating the most distinct descriptive difference in treatment-seeking behavior is family_history. Individuals with a family history of mental health issues have a noticeably higher rate of seeking treatment compared to those without such a history. One possible explanation is that individuals with a family history

of mental illness may have greater awareness of mental health conditions and available treatment options.

Another critical interaction appears between work_interfere and treatment. The overwhelming majority of employees who report that mental health issues "Often" or "Sometimes" affect their work life tend to utilize treatment mechanisms.

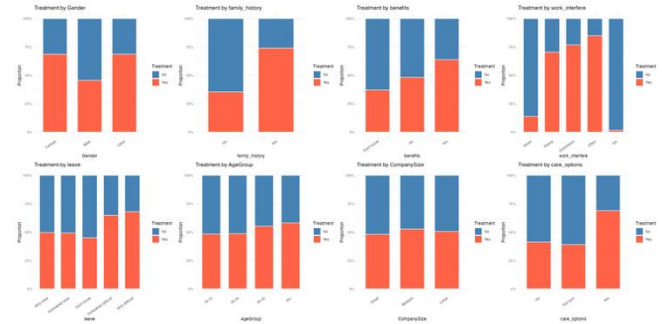


Figure 5: Bivariate Associations: Proportional Treatment Distribution by Categories

In contrast, the treatment-seeking rate is quite low among individuals who state that mental health issues "Never" affect their work life. This pattern suggests that individuals may be more likely to seek treatment when mental health difficulties begin to noticeably affect their daily work activities. At the corporate variable level, employees who know that their company explicitly provides mental health benefits (benefits = "Yes") show a higher tendency to seek treatment compared to those who are unaware of these rights or state that they are not provided.

IV. CONFIRMATORY DATA ANALYSIS

This section focuses on validating the patterns we observed during the exploratory data analysis. To do this, we used nominal and ordinal tests of independence, measured effect sizes using Odds Ratios and Relative Risks, and applied the Cochran-Mantel-Haenszel (CMH) test to control for potential confounding variables. We set the significance level (α) at 0.05 across all tests. A summary of the primary hypothesis tests is presented in Table 1.

Hypothesis	Test	Statistic	df	p_value	Decision
H1: family_history ~ treatment	Chi-square	175.2447	1	<0.001	Reject H0
H2: benefits ~ treatment	Chi-square	63.2397	2	<0.001	Reject H0
H3: work_interfere ~ treatment	Chi-square	293.0031	3	<0.001	Reject H0
MH: family_history ~ treatment Gender	Mantel-Haenszel	159.7229	1	<0.001	Reject H0

Table 1: Summary of Confirmatory Hypothesis Tests

A. Test of Independence

To examine how the target variable (treatment) relates to the nominal categorical predictors, we ran Pearson Chi-Square tests of independence. The standard hypotheses for these tests were set up as follows:

H0: There is no statistically significant relationship between mental health treatment-seeking behavior (treatment) and the independent variable.

H1: There is a statistically significant relationship between mental health treatment-seeking behavior and the independent variable.

The test results revealed that the p-values for family_history (chisq = 175.24, p < 0.001), work_interfere (chisq = 293.00, p < 0.001), and benefits (chisq = 63.24, p < 0.001) were all well below the 0.05 threshold. Consequently, we rejected the respective null hypotheses (H0) for these variables.

When we look at the risk-based effect sizes (Relative Risk and Odds Ratios), the forest plot in Figure 6 illustrates exactly how strongly these variables influence treatment-seeking behavior. The statistics confirm that individuals with a family history of mental health issues have a notably higher odds ratio of seeking treatment compared to those without such a history.

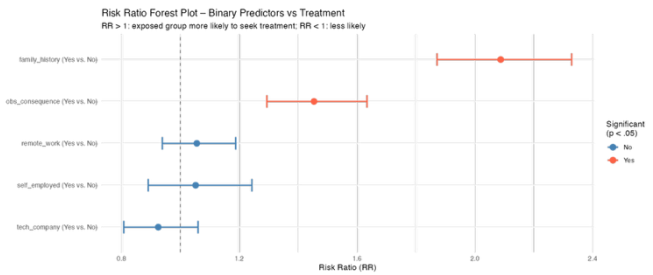


Figure 6: Forest Plot of Risk Ratios for Binary Predictors

B. Ordinal Associations Test

To measure the monotonic relationship between the ordinal variables, such as leave (ease of taking medical leave) and no_employees (company size), and the treatment variable, we used Kendall's Tau-b and Somers' D statistics.

H0: There is no significant monotonic relationship between the ordinal variable and treatment-seeking behavior.

H1: There is a significant monotonic relationship between the ordinal variable and treatment-seeking behavior.

According to the results, the number of employees (no_employees) does not have a sequential effect on seeking treatment. However, we found a statistically significant positive relationship between the leave variable, which represents the ease of taking leave for mental health issues, and treatment, leading us to reject the H0 hypothesis. The effect sizes for these ordinal variables are visualized in Figure 7.

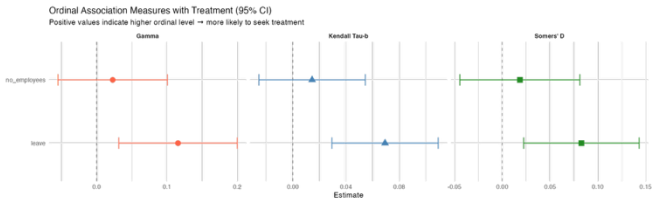


Figure 7: Ordinal Association Measures with Treatment (Kendall's Tau-b and Somers' D)

C. Conditional Independence

During the EDA and Chi-Square stages, we found that family_history and benefits were strong predictors. However, to test whether these relationships were affected by potential confounding variables like Gender and work_interfere, we applied the Cochran-Mantel-Haenszel (CMH) test using three-way contingency tables.

The first test was designed to measure the effect of family history on seeking treatment while controlling for the effect of gender:

H0: When Gender strata are held constant, treatment and family_history are conditionally independent.

H1: Even when controlling for Gender, there is a conditional relationship between treatment and family_history.

The CMH test yielded a Mantel-Haenszel Chi-Square statistic of 159.72 and a p-value < 0.001, leading to the rejection of the null hypothesis (H0). This confirms that even when accounting for the gender factor, family history has an independent and significant effect on seeking treatment. This stratified relationship is shown in Figure 8.

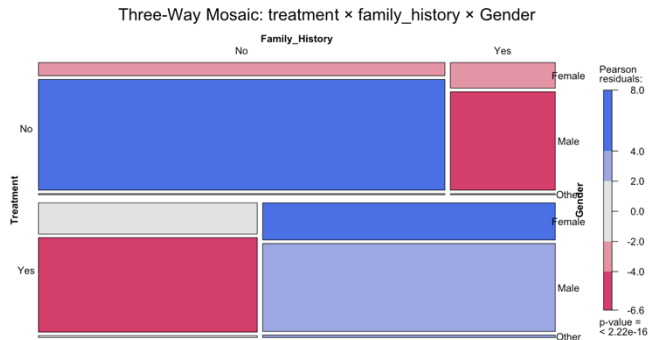


Figure 8: Three-way Mosaic Plot of Treatment and Family History Conditioned on Gender

Similarly, we tested the relationship between employer-provided rights (benefits) and treatment status by stratifying the level of work interference (work_interfere). This test also resulted in the rejection of H0, showing that employer-provided support increases the likelihood of seeking treatment, regardless of how much mental health issues interfere with work.

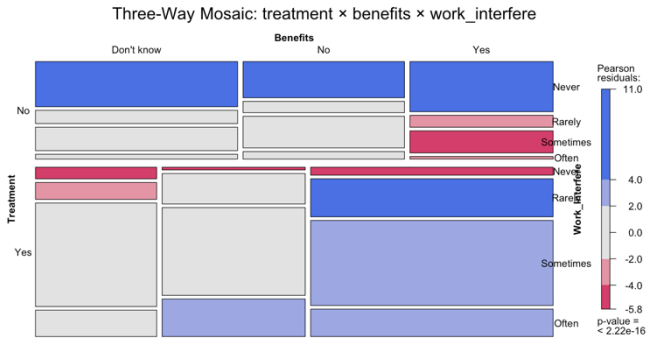


Figure 9: Three-way Mosaic Plot of Treatment and Benefits Conditioned on Work Interference

V. FEATURE ENGINEERING & DIMENSION REDUCTION

Following the confirmatory data analysis, the dataset underwent feature engineering to optimize it for machine learning algorithms. The main goal of this stage was to transform the variables in a way that boosts modeling performance while preserving statistical power.

First, to reduce the risk of overfitting caused by minor fluctuations in the continuous Age variable, we categorized it into AgeGroup. This transformation aligns with clinical and workforce dynamics, grouping individuals into five meaningful factor levels: 18-25, 26-35, 36-45, 46-55, and 55+.

Second, we reduced the dimensionality of the no_employees variable, which originally had too many levels. We recoded it as CompanySize to better reflect the practical distinction between small, medium, and large enterprises (e.g., 1-25 employees as "Small", 26-500 as "Medium", and 500+ as "Large"). Although the initial ordinal tests did not show a significant marginal effect for the raw employee count, this variable was recoded and retained for predictive modeling to capture potential multivariate interactions and to account for real-world corporate structure differences.

Additionally, we converted categorical variables into factor types and explicitly defined their reference categories using the relevel function in R, which is a strict requirement for regularization algorithms. As a result of these steps, the dataset was fully prepared for training the logistic and tree-based models detailed in Section VI.

VI. STATISTICAL MODELING (Prediction/Classification)

Building on the findings from the confirmatory analysis, we developed statistical and machine learning models to predict the likelihood of individuals seeking mental health treatment (predictive classification).

Before moving to the modeling phase, we split the dataset into 70% Training and 30% Testing subsets using the `set.seed(482)` argument. This approach allowed us to

objectively measure out-of-sample performance while strictly maintaining the balanced class distribution.

A. Parametric Classification Models (Logistic Regression)

Initially, we built a 'Full Model' (Full Logistic Regression) that included all independent predictors. Then, to improve model interpretability and reduce unnecessary complexity, we applied 'Stepwise' variable selection based on the Akaike Information Criterion (AIC).

The coefficients obtained from the stepwise logistic regression confirmed the direction of the predictor effects. Before proceeding with variable selection, we evaluated the Full Model's sensitivity to outliers using Cook's Distance, as visualized in Figure 10. We examined a few potential outliers but decided to keep them in the dataset, as we verified that they did not cause any major influential distortion on the model parameters.

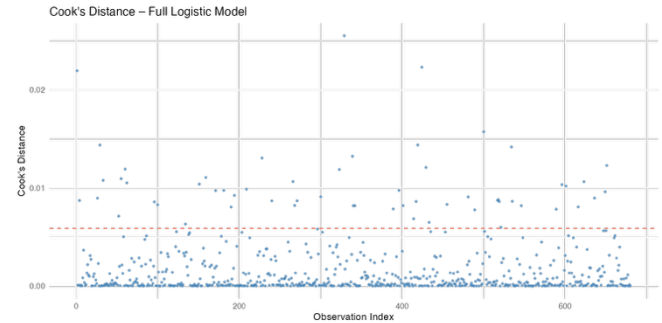


Figure 10: Cook's Distance Plot for Outlier Detection in the Logistic Regression Model

B. Regularized Logistic Regression (Ridge, Lasso, Elastic Net)

To address classical logistic regression's vulnerability to multicollinearity and to reduce prediction variance, we trained Regularization algorithms: Ridge (L2 penalty), Lasso (L1 penalty), and Elastic Net (L1+L2).

The penalty parameters (lambda) for these models were optimized based on the minimized deviance using 5-fold Cross-Validation. The Lasso model provided natural dimensionality reduction by shrinking some variable coefficients completely to zero, while Elastic Net both performed variable selection and kept coefficient magnitudes under control. The error curves from the cross-validation processes are presented in Figure 11.

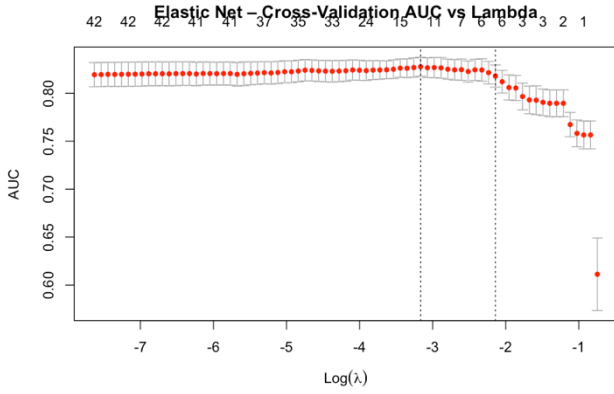


Figure 11: Cross-Validation AUC Curve for Elastic Net Tuning

C. Tree-Based Machine Learning Models

We also included tree-based models in our analysis, as they do not rely on parametric assumptions (like linearity) and can easily capture complex, non-linear interactions.

First, we constructed a Decision Tree model. However, to prevent overfitting, we pruned the tree using the Complexity Parameter (CP). We determined the optimal CP value using the 1-SE (Standard Error) rule to obtain the most stable tree structure (Figure 12).

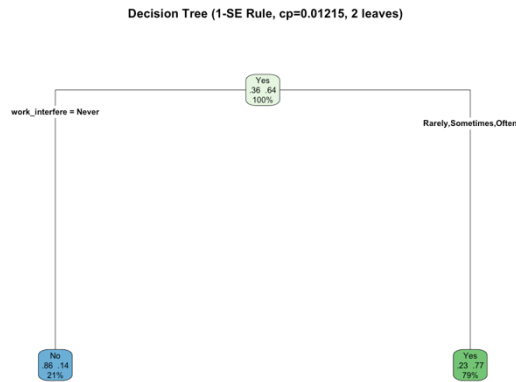


Figure 12: Pruned Decision Tree Structure using the 1-SE Rule

To maximize predictive power, we then utilized Random Forest, a much more robust ensemble learning method. We tuned the model's hyperparameters (mtry and ntree) again using cross-validation (mtry=7). The Variable Importance analysis, which is the biggest advantage of the Random Forest algorithm, proved once again that work_interfere and family_history are the top factors contributing to the model's overall accuracy (Figure 13).

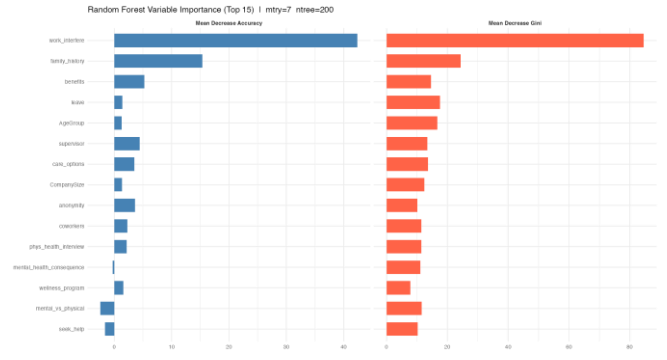


Figure 13: Random Forest Variable Importance (Mean Decrease Accuracy and Gini)

VII. MODEL EVALUATION & CONCLUSION

A. Model Performance Comparison

All algorithms developed in Section VI (Logistic Regression variations, Regularization models, and Tree-based models) were evaluated on the 30% hold-out test set, which was completely unseen during the training process. Since the class distribution of the target variable is balanced, Accuracy is considered a valid metric; however, the primary evaluation was based on Sensitivity, Specificity, F1-Score, and ROC-AUC (Area Under the Curve) metrics. The comparative test performance of these models is presented in Table 2 and Figure 14.

All metrics computed on held-out 30% test set (set.seed = 482)

Model	Classification Metrics					Discrimination	
	Accuracy	Sensitivity	Specificity	Precision	Recall	F1	ROC_AUC
Logistic Full	0.7690	0.8757	0.5810	0.7864	0.8757	0.8286	0.8028
Logistic Selected	0.7793	0.8919	0.5810	0.7895	0.8919	0.8376	0.8059
Ridge	0.7828	0.9568	0.4762	0.7629	0.9568	0.8489	0.8057
Lasso	0.7793	0.9135	0.5429	0.7788	0.9135	0.8408	0.8056
Elastic Net	0.7793	0.9189	0.5333	0.7763	0.9189	0.8416	0.8067
Decision Tree	0.7828	0.9514	0.4857	0.7652	0.9514	0.8482	0.7185
Random Forest	0.7552	0.8757	0.5429	0.7714	0.8757	0.8203	0.8043

Best model (highest AUC) highlighted in green.

Table 2: Model Performance Comparison on Test Set

When we look at the test results, we observed that the classical "Full" logistic regression model (AUC = 0.8028) slightly improved its performance after applying Stepwise variable selection (AUC = 0.8059). However, among the machine learning approaches, the Elastic Net model stands out by simultaneously optimizing model complexity and handling multicollinearity issues. With a high discriminative power (ROC-AUC = 0.8067) and a Sensitivity of 91.89%, Elastic Net achieved the best overall balance among the evaluated models in identifying individuals who reported seeking treatment. Although the Random Forest algorithm also offers strong predictive potential, we selected Elastic Net as the final champion model because of the transparency of its coefficients and the clear interpretability of its effect directions (Odds Ratios).

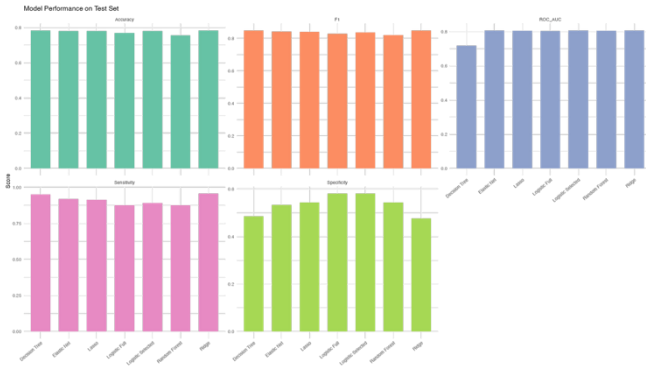


Figure 14: Visual Comparison of Classification Metrics Across Evaluated Models

B. Conclusion and Practical Implications

The statistical inference and predictive modeling findings of this study show that the mental health treatment-seeking behavior of tech sector employees is shaped by a combination of both individual and corporate factors.

The estimated effects and variable importance rankings are generally consistent with the patterns observed in this dataset. As mental health issues increasingly interfere with daily functioning and work (work_interfere), the odds of receiving professional help rise dramatically.

Similarly, family history of mental illness emerged as one of the strongest demographic factors associated with treatment-seeking behavior. On the corporate side, employer-provided mental health benefits (benefits) and the ease of taking medical leave (leave) create a positive, encouraging effect on the treatment process. These findings suggest that workplace policies and support mechanisms are closely related to treatment-seeking behavior among employees. Based on these findings, organizations may benefit from improving access to mental health resources, strengthening confidentiality practices, and communicating employee support programs more clearly.

C. Limitations

The validity of this analysis is bounded by certain assumptions and data-driven limitations. First, because the dataset is cross-sectional, the relationships identified between variables represent strong correlations and predictive power, not causality. Second, for practical modeling purposes, we treated the missing observations as non-informative and handled them using mode imputation. The potential bias this approach might introduce to the model variance could be tested using more complex techniques like multiple imputation in larger-scale studies. Finally, since the dataset was collected from self-selected survey participants, the generalizability of the findings to broader populations outside the tech sector may be limited.

ACKNOWLEDGMENT

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